CHANDRA

Sr. Big Data Engineer | 10+ Years Experience

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OBJECTIVE / SUMMARY

Results-driven Sr. Big Data Engineer with 10+ years of experience designing and deploying production-grade data pipelines, agentic AI/LLM systems, and cloud-native architectures across healthcare, finance, aviation, and technology sectors. Expert in Python, Apache Spark, AWS, Azure, Dagster, DSPy, LangChain, and distributed systems. Proven track record of leading complex migrations, building HIPAA/PCI-DSS/SOX-compliant platforms, and delivering scalable solutions processing millions of records daily. Deep expertise in LLM prompt optimization, NER-based metadata classification, and real-time streaming architectures.

TECHNICAL SKILLS

Languages: Python, Scala, Java, PySpark, SQL, T-SQL, Unix, Spring Boot

Cloud: AWS (S3, Glue, EMR, Lambda, ECS, EKS, EC2, RDS, Redshift, SageMaker, CDK, EventBridge, CloudWatch, SQS, SNS), GCP (BigQuery, DataProc, GKE, Spanner, Cloud Functions), Azure (Data Factory, Databricks, SQL, Data Lake, Blob Storage, HDInsight, CosmosDB)

Big Data: Apache Spark, Kafka, Airflow, Hive, HDFS, NiFi, Spark Streaming, Zookeeper, Sqoop, Flume, MapReduce, Yarn, Dagster

Databases: Teradata 15/14, MySQL, Oracle 11g/10g, MS SQL Server 2014/2012, MongoDB, PostgreSQL, Snowflake

ETL Tools: AWS Glue, Informatica PowerCenter 10.x/9.6, Talend 5.6, SSIS, Semarchy xDM

ML/AI: DSPy, LangChain, LangGraph, MLflow, MLLIB, NumPy, Pandas, Scikit-Learn, Gradient Boosting, Random Forest, Logistic Regression

DevOps: Docker, Kubernetes, Jenkins, Git, AWS CDK, SonarQube, PMD

BI Tools: Tableau, Power BI, Grafana, Prometheus, Collibra, Splunk

Compliance: Agile, Waterfall, SDLC, HIPAA, HITECH, PCI-DSS, SOX, GLBA, SOC 2, FAA/IATA

WORK EXPERIENCE

Humana - Sr. Big Data Engineer

August 2023 – Present | Louisville, KY

- Designed and deployed agentic retrieval workflows using Python, LangChain, and LangGraph to build context-aware agents ingesting PR metadata, commit metadata, and GitHub history for automated CES scoring and healthcare quality metrics.
- Migrated legacy orchestration from Apache Airflow to Dagster across 40+ production pipelines processing healthcare claims, clinical records, and patient data on AWS ECS and AWS EKS with retries, backfills, and data quality alerts.
- Built production-grade LLM-driven scoring systems using DSPy, integrating prompt optimization, chaining, and context packing to automate healthcare PR-level scoring using GPT-4, Claude, AWS SageMaker, and MLflow.
- Engineered backend services using Python, AWS Lambda, AWS EventBridge, and AWS CDK to ingest PR and commit metadata from GitHub and Jenkins, computing CES metrics and quality signals via RESTful APIs and Power BI dashboards.
- Developed data modeling frameworks using Pandas, NumPy, and MLLIB to link PRs, commits, branches, and epics for healthcare software delivery metrics with graph-based relationships across Jira, GitHub, and AWS S3.
- Orchestrated large-scale backfills and historical corrections using Dagster and AWS Glue, processing over 10 million commits and clinical record deltas with batching, caching, and DSPy-based score optimization.
- Implemented SonarQube integration for code-quality scoring, linking static analysis results with GitHub PR metadata and commit history to enrich CES metrics ensuring HIPAA-compliant metadata governance.
- Designed eval and feedback loops using Python, DSPy, and MLflow for LLM-driven prompt optimization creating versioned evaluation datasets and collecting feedback via Power BI and Jira.
- Built observability and monitoring solutions using AWS CloudWatch, AWS Lambda, and Dagster asset sensors to track metrics, traces, logs, and data quality alerts across healthcare ETL pipelines.
- Deployed AWS CDK infrastructure for pipelines on AWS EC2, AWS EKS, AWS Lambda, and AWS S3, automating provisioning, monitoring, and scaling of Dagster orchestrators and LLM inference endpoints.
- Developed and deployed AI/ML pipelines using MLLIB and NumPy to predict patient readmission risk, orchestrated with Airflow ensuring HIPAA-compliant data handling and HL7 interoperability.
- Implemented conversational AI chatbots using LLMs and AWS EC2 to automate patient triage and scheduling, integrating with EHR systems through secure API gateways and access logging.

- Environment: AWS SageMaker, Dagster, DSPy, LangChain, LangGraph, Python, MLflow, Power BI, Collibra, ERWIN, Pandas, NumPy, MLLIB, AWS EC2, AWS RDS, S3, Lambda, Jenkins, EventBridge, Git, GitHub, AWS EKS, Glue, Airflow, ECS, SonarQube

Morgan Stanley - Sr. Data Engineer

October 2020 – July 2023 | New York, NY

- Designed and deployed production-grade data pipelines using Python and Dagster on AWS, ingesting PR metadata, commit metadata, and code quality telemetry from GitHub and Jenkins ensuring PCI-DSS-compliant financial workflows.
- Built retrieval agents using Python and DSPy to extract historical code context, diffs, and commit histories from GitHub, implementing prompt chaining and context packing for LLM optimization with SOX-compliant audit trails.
- Migrated orchestration from Apache Airflow to Dagster, enhancing PR/commit linking, branch tracking, and epic linking across Jira, GitHub, and banking systems with PCI-DSS and SOX compliance.
- Developed anomaly detection and data quality monitoring using Python, AWS Glue, and AWS CloudWatch, identifying deviations in commit metadata and code quality signals from SonarQube and PMD.
- Engineered AWS-native pipelines using AWS ECS, AWS Lambda, AWS S3, AWS RDS, and AWS CDK to automate backfills, retries, and scheduling for high-volume financial data ingestion with PCI-DSS encryption.
- Implemented observability and monitoring frameworks using AWS CloudWatch, AWS X-Ray, and Python scripts tracking metrics, traces, and logs ensuring PCI-DSS, SOX, and GLBA compliance.
- Built backend services using Python, AWS Lambda, and AWS EKS to ingest PR metadata, commit metadata, and code quality signals computing CES and quality metrics via JWT-authenticated REST APIs.
- Designed and executed DSPy-based eval sets and feedback loops, collecting user feedback on scoring accuracy, retraining LLM/prompt optimization models with batching and caching for GLBA-compliant workflows.
- Implemented AWS WAF rules and security policies for AWS EC2-hosted banking applications, mitigating SQL injection, XSS, and DDoS attacks ensuring GLBA compliance and secure customer authentication.
- Developed infrastructure-as-code templates using AWS CDK in TypeScript to provision AWS EC2, VPCs, security groups, and IAM roles for loan origination systems ensuring PCI-DSS and SOX compliance.
- Configured Airflow DAGs to automate inter-bank settlement reconciliations orchestrating validation, anomaly detection, and report generation with SOX compliance and SLA adherence.
- Environment: Lambda, Jenkins, Airflow, S3, SonarQube, Dagster, ERWIN, JWT, ECS, AWS EC2, AWS RDS, Power BI, AWS EKS, AWS WAF, Jira, AWS CDK, DSPy, Python, Glue, Git, PMD, GitHub, EventBridge, AWS SQS

Delta Airlines - Data Engineer / Data Analyst

December 2017 – September 2020 | Atlanta, GA

- Designed and deployed production-grade data pipelines using Python and Apache Airflow to ingest commit metadata and PR metadata from GitHub integrating AWS Lambda and AWS EventBridge for real-time processing with FAA/IATA compliance.
- Built end-to-end data modeling frameworks using Python to link PRs and commits, tracking branch merges to main repositories, integrating AWS Glue and Dagster pipelines for historical backfills and IOSA-compliant aviation software quality monitoring.
- Engineered retrieval agents using Python to extract contextual metadata from GitHub, integrating SonarQube outputs into AWS S3 data lakes and computing technical debt and maintainability metrics.
- Orchestrated migration of legacy workflows using Dagster and Airflow on AWS ECS and AWS EKS, implementing retry logic, scheduling policies, and monitoring frameworks with Grafana and AWS SNS notifications.
- Developed analytics platforms using Python, AWS RDS, and Power BI to generate CES metrics, correlating PR and commit metadata with Jira epics and designing schemas in ERWIN and E/R Studio.
- Built observability frameworks using Prometheus, Grafana, and AWS CloudWatch tracking Airflow DAG execution times, AWS Glue job success rates, and Lambda invocation patterns for SOC 2 compliance.
- Integrated SonarQube with Airflow and AWS Glue, ingesting static analysis metrics into AWS S3, mapping code severity to FAA DAL levels and implementing automated risk scoring.
- Built analytics dashboards in Power BI connected to AWS RDS and AWS S3 computing trends in code review turnaround, defect escape rates, and test coverage with Collibra for data catalog governance.
- Environment: AWS CDK, Python, ERWIN, AWS EC2, EventBridge, GitHub, SonarQube, PMD, AWS Lambda, Prometheus, AWS EKS, Dagster, Airflow, DSPy, AWS WAF, AWS S3, ECS, Glue, Jenkins, AWS RDS, Grafana, Power BI, Git, AWS SQS, Collibra, AWS SNS, Jira

Cisco - Data Engineer / Migration Specialist

February 2016 – November 2017 | San Jose, CA

- Built a Network Data Integration platform using Azure Cloud, improving enterprise data integration efficiency and enabling seamless data flow across global networking systems with monitoring telemetry from thousands of devices.
- Developed scalable data ingestion pipelines using Talend ETL and Azure Data Factory, efficiently processing large volumes of network monitoring and security appliance data.
- Executed data migration initiatives transferring configuration data from legacy network management systems to modern SDN platforms, maintaining configuration accuracy and uninterrupted network services.
- Built complex orchestration workflows using Apache Airflow, coordinating ETL processes across distributed network monitoring systems in multiple global regions.
- Implemented Azure SQL Database to store structured network performance metrics, optimizing data models and query execution for rapid access to critical network health indicators.
- Developed predictive capacity forecasting models using Azure Databricks, analyzing historical traffic trends to improve bandwidth planning and support proactive decision making.
- Created real-time network anomaly detection solutions using streaming analytics, enabling early identification of security threats and performance degradation.
- Environment: Azure Cloud, Talend ETL, Azure Data Factory, Apache Airflow, Azure SQL Database, Azure Data Lake Storage, Azure Databricks, Python, Streaming Analytics, SDN Platforms

Flipkart - ETL Developer

August 2014 – December 2015 | Bangalore, India

- Built a scalable ETL pipeline using AWS Services, Informatica, and Talend, integrating data from e-commerce platforms, inventory systems, and customer databases for high-volume daily transaction processing.
- Utilized Informatica PowerCenter to extract data from MySQL, PostgreSQL, and MongoDB sources containing customer profiles and purchase history ensuring accurate extraction of high-volume records.
- Designed AWS Glue jobs to catalog, transform, and load curated sales and inventory data into Amazon Redshift, optimizing storage layouts and enabling efficient analytical queries.
- Automated ETL job scheduling using AWS Lambda and Amazon CloudWatch Events, improving reliability of daily sales reporting and inventory refreshes.
- Connected Tableau to Amazon Redshift to deliver interactive dashboards for real-time analysis of sales trends, inventory positions, and customer behavior.
- Environment: AWS (S3, Glue, Lambda, CloudWatch, Redshift), Informatica PowerCenter, Talend, MySQL, PostgreSQL, MongoDB, Tableau

EDUCATION

Bachelor of Technology in Computer Science & Engineering | Koneru Lakshmaiah University | 2010–2014

KEY ACHIEVEMENTS & METRICS

- Migrated 40+ production pipelines from Apache Airflow to Dagster at Humana, processing healthcare claims with 99.9%+ reliability.
- Processed 10+ million commits and clinical record deltas using Dagster and AWS Glue with batching and DSPy-based score optimization.
- Designed HIPAA/HITECH/PCI-DSS/SOX-compliant data architectures across healthcare and financial services for Fortune 500 clients.
- Built agentic LLM systems using DSPy, LangChain, and LangGraph reducing manual code review effort by automating CES metric computation.
- Deployed multi-cloud architectures spanning AWS, GCP, and Azure for real-time and batch data processing at petabyte scale.
- Led data engineering teams through complex ETL migrations, cloud modernization programs, and ML pipeline deployments.
- Integrated SonarQube, PMD, and GitHub code quality telemetry into automated scoring pipelines improving code quality visibility.
- Established observability standards with Prometheus, Grafana, AWS CloudWatch achieving SLA adherence across aviation safety systems.